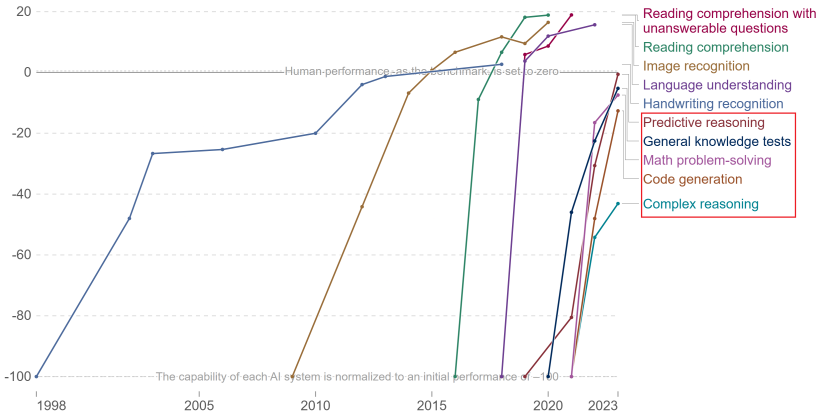
Graded Representation Theory of Equivariant Neural Networks

Mani Shayestehfar

Supervised by Dani Tubbenhauer

Test scores of AI systems on various capabilities relative to human performance

Our World in Data

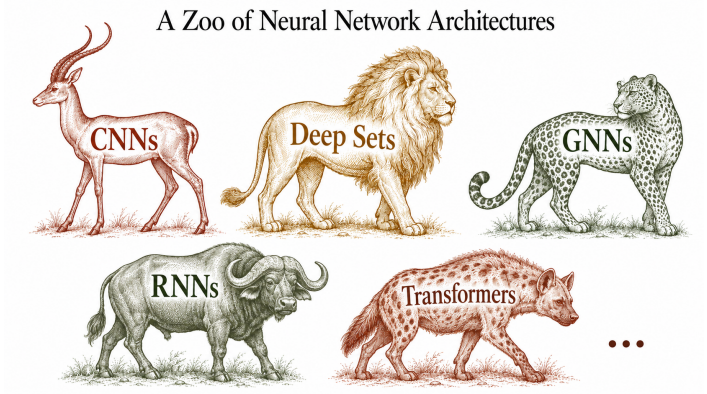


Data source: Kiela et al. (2023)

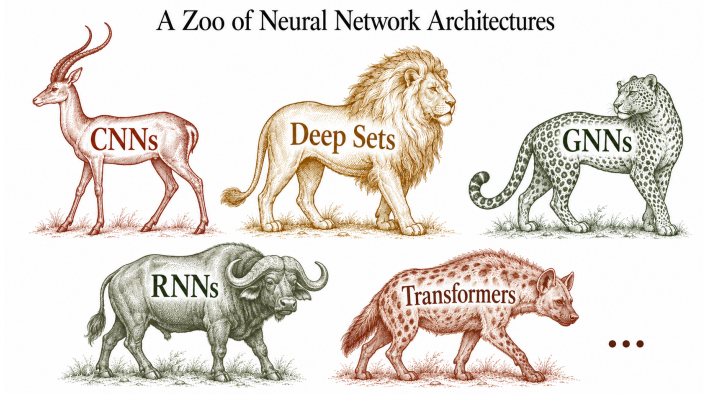
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THE MISSING LINK

There is no single mathematical framework that completely explains the behaviour of neural networks.



Geometric Deep Learning: Symmetry provides a common organising principle.



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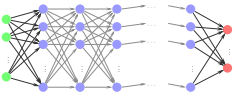
Symmetry in data $\equiv$ action on some abstract vector space

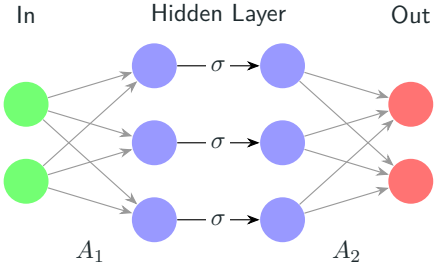
Inspired by ShutterStock image used in GDL

Equivariant **Neural Networks**

$\sigma: \mathbb{R} \to \mathbb{R}$ Piecewise-linear

Weights A_1, A_2 take weighted sums; σ is the *activation*.

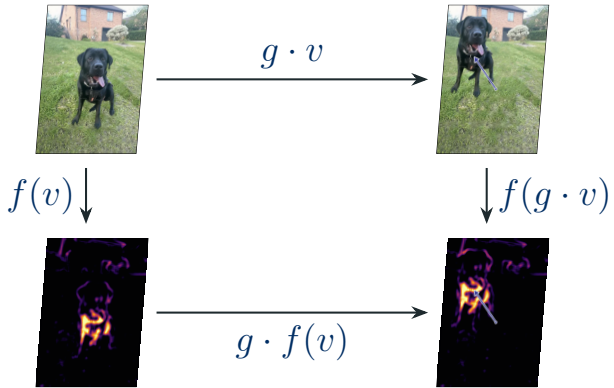




$$f_w(v) = A_2\Big(\sigma(A_1(v))\Big), \qquad w = (A_1, A_2) \in W.$$

A fixed architecture defines a family: $\mathcal{F} = \{f_w : w \in W\}$.

Equivariant Neural Networks (Interactive toy)



$$f(g \cdot v) = g \cdot f(v)$$

Our Setting

Representation G finite, $\rho_V : G \to GL(V)$, V a $\mathbb{R}$-Vector Space

Permutation rep. G permutes the basis $\{\delta_b\}_{b \in B}$ of Func$(B, \mathbb{R})$: $g \cdot \delta_b = \delta_{gb}$

e.g. $(1\,2\,3) \cdot (v_1, v_2, v_3) = (v_3, v_1, v_2)$

Irreducible no smaller nonzero subspace preserved by G

Equivariant map $f(g \cdot v) = g \cdot f(v)$; linear ones: $\mathrm{Hom}_G(V, W)$

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Irreducible	no smaller nonzero subspace preserved by G
Equivariant map	$f(g \cdot v) = g \cdot f(v)$; linear ones: $\mathrm{Hom}_G(V, W)$

The network, algebraically

$$V_{\text{in}} \xrightarrow{A_1} H \xrightarrow{\sigma} H \xrightarrow{A_2} V_{\text{out}}, \qquad H = \text{Func}(B, \mathbb{R}), \qquad A_i \in \mathrm{Hom}_G\,.$$

σ acts on each coordinate; G only **shuffles** them $\Rightarrow$ σ is equivariant.

What changes for nonlinear maps?

Let $L \ncong K$ be irreducible and $f : L \to K$ equivariant.

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Weights never mix irreducible types.

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Nonlinear interaction

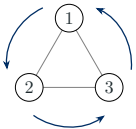
For irreducible summands L, K of the hidden layer H:

$$f_L^K : \ L \xhookrightarrow{\iota_L} H \xrightarrow{\sigma} H \xrightarrow{p_K} K$$

is equivariant (as σ is), and **may be nonzero**.

An example

S_3 permutes the coordinates of

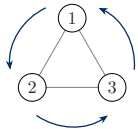


$$\mathbb{R}^3 = \underbrace{\mathbb{R}(1,1,1)}_{L_0\text{: constant}} \oplus \underbrace{\{x_1 + x_2 + x_3 = 0\}}_{L_1\text{: zero sum}}.$$

For example, $(1,2,3) = (2,2,2) + (-1,0,1)$: average plus deviations.

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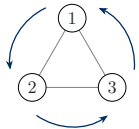
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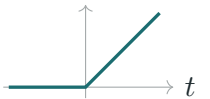


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Linear: $\quad \mathrm{Hom}_{S_3}(L_1, L_0) = 0.$

Nonlinear:



$\mathrm{ReLU}(t) = \max(t, 0)$

$$(-1,0,1) \xmapsto{\text{ReLU}} (0,0,1) \xmapsto{p_0 \text{ (average)}} \tfrac{1}{3}(1,1,1) \neq 0.$$

An equivariant map $L_1 \to L_0$ exists, but only nonlinearly.

When can a nonlinear interaction exist?

PL Schur's Lemma (Gibson–Tubbenhauer–Williamson)

Let G be finite and L, K irreducible real representations. Then

$$\mathrm{Hom}_G^{\mathrm{PL}}(L, K) \neq 0 \quad \Longleftrightarrow \quad \ker \rho_L \subseteq \ker \rho_K.$$

- $\mathrm{Hom}_G^{\mathrm{PL}}$: equivariant, continuous, affine on finitely many polyhedra.
- $\ker \rho_L$: normal subgroup of G, $\rho_L : G \to GL(L)$
- In words: every symmetry acting trivially on L must also act trivially on K.

S_3: $\ker \rho_{L_1} = \{1\} \subseteq \ker \rho_{L_0} = S_3$, so $L_1 \to L_0$ is possible; $L_0 \to L_1$ is not.

What does this thesis add?

$$\mathrm{Hom}_G^{\mathrm{PL}}(L, K) \neq 0 \iff \ker \rho_L \subseteq \ker \rho_K$$

GTW: *whether* a nonlinear interaction can exist.

This thesis: characterise nonlinear interactions, by *"degree"*.

What do we mean by degree?

Think linear, quadratic, cubic: t, t^2, t^3.

We will extend this idea to nonlinear activations such as ReLU.

Why can an interaction be missing?

For one layer:

$$F : L \xrightarrow{A_1} H \xrightarrow{\sigma} H \xrightarrow{A_2} K.$$

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Symmetry	forbids it,
Coordinates	(A_1, A_2) miss it,
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Main message

Degree by degree, these three causes can be **separated**.

Rescuing Schur's lemma

A homogeneous degree-k polynomial is **linear** in $x \otimes \cdots \otimes x$:

$$\|x\|^2 = \sum_i x_i x_i \quad \text{is linear in } x \otimes x.$$

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These tensors span $\operatorname{Sym}^k L$, the symmetric kth tensor power. Equivariant degree-k polynomial maps $L \to K$ correspond to

$$\operatorname{Hom}_G(\operatorname{Sym}^k L, K).$$

Schur's lemma applies again: K must occur in $\operatorname{Sym}^k L$.

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S_3 example

$\operatorname{Sym}^2 L_1 \cong L_0 \oplus L_1$ contains L_0.

Hence $x \mapsto \|x\|^2(1,1,1)$ gives an equivariant map $L_1 \to L_0$.

What **symmetry** permits

- GTW tells us *whether* an interaction can exist.
- We now ask **which polynomial degrees** permit it.
- Write $D(L, K) = \{k \geq 0 \,:\, \mathrm{Hom}_G(\mathrm{Sym}^k L, K) \neq 0\}$.

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Theorem (degree detection) (M.S.)

For $L \neq 0$ and K irreducible:

$$D(L,K) \neq \varnothing \iff \ker \rho_L \subseteq \ker \rho_K.$$

$$S_3: \quad D(L_1,L_0) = \{0,\cancel{1},2,3,\ldots\} \qquad \underbrace{\|x\|^2(1,1,1)}_{\text{degree 2}},\ \underbrace{x_1x_2x_3(1,1,1)}_{\text{degree 3}}$$

How can ReLU have a degree?

ReLU is not a polynomial, yet we still want its **degree parts**.

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ReLU is not a polynomial, yet we still want its **degree parts**.

Find the right **embedding**. With the Gaussian inner product

$$\langle f, g \rangle_{L^2(\gamma_{\mathbb{R}})} := \int_{\mathbb{R}} f(t)\, g(t)\, d\gamma_{\mathbb{R}}(t),$$

split maps orthogonally, in **Hermite basis**:

$$L^2(\gamma_{\mathbb{R}}) = \widehat{\bigoplus_{k \geq 0}} \mathbb{R}\, \mathrm{He}_k, \qquad \mathrm{He}_k = t^k - \Big(\text{projection of } t^k \text{ onto lower degrees}\Big).$$

$$\mathrm{He}_0 = 1, \quad \mathrm{He}_1 = t, \quad \mathrm{He}_2 = t^2 - 1, \quad \mathrm{He}_3 = t^3 - 3t, \quad \dots$$

How can ReLU have a degree?

Degree-k part of f $:=$ orthogonal projection of f onto $\mathbb{R}\,\mathrm{He}_k$.

ReLU has no cubic part

$$\mathrm{ReLU}(t) = c\cdot 1 \;+\; \tfrac{1}{2}t \;+\; \tfrac{c}{2}\,\mathrm{He}_2(t) \;+\; 0\cdot\mathrm{He}_3(t) \;-\; \tfrac{c}{24}\,\mathrm{He}_4(t) \;+\; \cdots, \qquad c = \tfrac{1}{\sqrt{2\pi}}.$$

Why: $\mathrm{ReLU}(t) = \frac{t}{2} + \frac{|t|}{2}$; $|t|$ is even and $\frac{t}{2}$ is linear, so both are $\perp \mathrm{He}_3$.

We can now ask **which degrees an activation keeps**.

A degree for equivariant maps

The same grading works on L (with a G-invariant inner product):

$$L^2(\gamma_L) = \widehat{\bigoplus_{k\geq 0}} \mathcal{H}_k(L), \qquad \mathcal{H}_k(L) \cong \operatorname{Sym}^k L \text{ as } G\text{-representations.}$$

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- The **whole layer** splits: $F = \sum_{k\geq 0} \Pi_k F$, for $F \in L^2(\gamma_L; K)$.
- γ_L is G-invariant, so each degree part $\Pi_k F$ is still **equivariant**.

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Theorem (graded decomposition) (semi-classical)

$$L^2(\gamma_L; K)^G \cong \widehat{\bigoplus_{k\geq 0}} \operatorname{Hom}_G(\operatorname{Sym}^k L, K).$$

The degree-k part of any layer lives in $\operatorname{Hom}_G(\operatorname{Sym}^k L, K)$, the space from "Rescuing Schur's lemma".

Main theorem

$F(x) = A_2\,\sigma(A_1 x),\;\; H = \mathsf{Func}(B, \mathbb{R})$ with G transitive on B.

- **Coordinates** produce a candidate degree-k interaction. Replacing σ by $t \mapsto t^k$ (in each coordinate) gives

$$Q_k(x) = A_2\Big((A_1 x)^{\odot k}\Big) = \sum_{b \in B} \langle x, w_b \rangle^k\, A_2 \delta_b, \qquad w_b := A_1^* \delta_b.$$

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- **Activation** keeps or removes it through its Hermite coefficient

$$a_k(\sigma; s) = \tfrac{1}{k!} \left\langle \sigma_s, \mathrm{He}_k \right\rangle_{L^2(\gamma_{\mathbb{R}})}, \qquad \sigma_s(t) := \sigma(st).$$

$s > 0$: the common length $\|a_b\|$ of the rows of A_1 (equal for all b, as G is transitive on B).

Main theorem

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- **Coordinates:** $Q_k(x) = A_2\Big((A_1 x)^{\odot k}\Big)$.
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Theorem (coordinatewise factorisation) (M.S.)

$$\Pi_k F = \frac{a_k(\sigma; s)}{s^k}\,\Pi_k Q_k.$$

Equivalently, $\Pi_k F \neq 0$ iff ($Q_k \neq 0$ and $a_k \neq 0$).

Idea: in each hidden coordinate, expand $\sigma(st) = \sum_j a_j(\sigma; s)\,\mathrm{He}_j(t)$; only the $j = k$ term has a degree-k part.

S_3: ReLU kills the cubic

$$L_1 \hookrightarrow \mathbb{R}^3 \xrightarrow{\text{ReLU}} \mathbb{R}^3 \xrightarrow{p_0} L_0.$$

- **Symmetry** allows a cubic interaction: $3 \in D(L_1, L_0)$.

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- **Symmetry** allows a cubic interaction: $3 \in D(L_1, L_0)$.
- **Coordinates** produce it: $Q_3(x) = x_1 x_2 x_3 (1,1,1) \neq 0$.
- **ReLU** removes it: its cubic coefficient is zero.

$$\text{ReLU}(t) = \tfrac{t}{2} + \tfrac{|t|}{2} \quad \Longrightarrow \quad a_3 = 0.$$

The activation can remove an interaction allowed by symmetry.

What should we remember?

Is degree $k\ldots$	Test	Fix
permitted?	$\mathrm{Hom}_G(\mathrm{Sym}^k L, K) \neq 0$	change L or K
produced?	$Q_k \neq 0$	change A_1, A_2
kept?	$a_k(\sigma; s) \neq 0$	change σ or bias

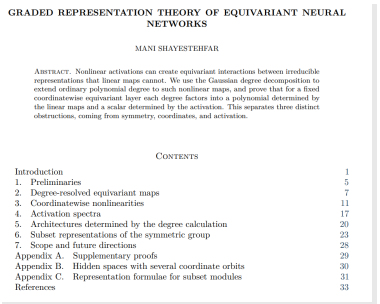
- **GTW:** *whether* a nonlinear interaction can exist.
- **This work:** in *which degree*, and *why* a layer realises it or not.
- **Clopen Problems:** Applications? other symmetries? which chaos sequences come from PL maps?

Preprint and workshop paper

Preprint available arXiv:2609.25776

Accepted to the NeurIPS 2026 Workshop

Symmetry and Geometry in Neural Representations



GRADED REPRESENTATION THEORY OF EQUIVARIANT NEURAL NETWORKS

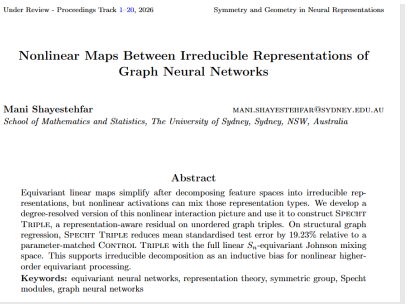
MANI SHAYESTEHFAR

ABSTRACT. Nonlinear activations can create equivariant interactions between irreducible representations that linear maps cannot. We use the Gaussian degree decomposition to extend ordinary polynomial degree to such nonlinear maps, and prove that for a fixed coordinatewise equivariant layer each degree factors into a polynomial determined by the linear maps and a scalar determined by the activation. This separates three distinct obstructions, coming from symmetry, coordinates, and activation.

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Preprint



Nonlinear Maps Between Irreducible Representations of Graph Neural Networks

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Abstract

Equivariant linear maps simplify after decomposing feature spaces into irreducible representations, but nonlinear activations can mix those representation types. We develop a degree-resolved version of this nonlinear interaction picture and use it to construct SPECHT TRIPLE, a representation-aware residual on unordered graph triples. On structural graph regression, SPECHT TRIPLE reduces mean standardised test error by 19.23% relative to a parameter-matched CONTROL TRIPLE with the full linear S_n-equivariant Johnson mixing space. This supports irreducible decomposition as an inductive bias for nonlinear higher-order equivariant processing.

Keywords: equivariant neural networks, representation theory, symmetric group, Specht modules, graph neural networks

Workshop paper

I would also like to thank my supervisor, Dani Tubbenhauer, for their patience and guidance throughout!

References

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[Sha26a] M. Shayestehfar. *Graded representation theory of equivariant neural networks.* arXiv:2609.25776, 2026.

[Sha26b] M. Shayestehfar. *Nonlinear maps between irreducible representations in graph neural networks.* NeurIPS 2026 Workshop on Symmetry and Geometry in Neural Representations.

Figures: test scores from Our World in Data (data: Kiela et al., 2023); architecture zoo inspired by an image used in [BBCV21].

S_3: ReLU kills the cubic

- Keep the same coordinates and **add a bias** to ReLU.
- A nonzero bias gives a **nonzero cubic coefficient**.

Fix: Shifted ReLU

$$\sigma_c(t) = \mathrm{ReLU}(t - c).$$

$$a_3(\sigma_c; s) = \tfrac{c}{6}\varphi(c/s) \neq 0 \text{ for } c \neq 0.$$

φ is the standard Gaussian density.

The cubic interaction now appears.

Why the Gaussian measure γ_L?

- **Symmetry:** it is $O(L)$-invariant, so the parts stay G-equivariant.
- **Completeness:** polynomials are dense in $L^2(\gamma_L)$, so the parts reconstruct the map.
- **Independence:** coordinates are independent in every orthonormal basis. By Herschel–Maxwell ($\dim L \geq 2$), this characterises centred isotropic Gaussians.

Theorem (Herschel–Maxwell)

Let μ be a probability measure on $\mathbb{R}^n$, $n \geq 2$. If μ is $O(n)$-invariant and is the product of its one-dimensional marginals, then

$$\mu = N(0, \sigma^2 I_n) \quad \text{for some } \sigma \geq 0.$$